

Rudranil Mondal

Computer Science Undergraduate — Aspiring Software Engineer

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EDUCATION

National Institute of Technology, Durgapur

West Bengal, India

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.20

Aug. 2023 – May 2027

Sarada Vidyapith (WBCHSE)

Kolkata, West Bengal, India

Higher Secondary (Class XII) — 92.4%

2023

Sarada Vidyapith (WBBSE)

Kolkata, West Bengal, India

Secondary (Class X) — 96.14%

2021

PROJECTS

Multi-Agent Orchestration System *Feb 2026 – May 2026* | [GitHub](#) | [Live Demo](#)

Python, FastAPI, LangGraph, LangChain, React, TanStack Start, SSE

- Built a real-time multi-agent orchestration system where specialized agents collaborate using a supervisor-based routing architecture.
- Implemented LLM-driven task decomposition and dynamic agent routing using LangGraph state management.
- Designed SSE-based streaming architecture to visualize live agent execution in a React dashboard.
- Integrated tools for web search, Python execution, and document generation enabling autonomous workflows.

AI Mock Interviewer (Voice-Based) *Apr 2025 – Jul 2025* | [GitHub](#) | [Live Demo](#)

Next.js, Vapi, OpenAI, Deepgram, Firebase, Tailwind

- Built a full-stack voice-based AI mock interview platform that simulates technical and behavioral interviews using conversational AI.
- Integrated Vapi, Deepgram, and GROQ APIs to enable real-time speech recognition, text-to-speech, and dynamic interview question generation.
- Implemented an AI-powered feedback pipeline that analyzes interview transcripts and generates structured evaluations across communication, technical knowledge, and problem-solving skills.
- Developed secure authentication and interview management using Firebase, enabling persistent interview history and reusable interview sessions.

AI Study Assistant *Nov 2025 – Jan 2026* | [GitHub](#) | [Live Demo](#)

React, FastAPI, Transformers, OCR, HuggingFace, Python

- Built an AI-powered study assistant that transforms PDFs, images, and raw text into structured explanations using OCR, semantic retrieval, and transformer-based LLMs.
- Designed a semantic chunking, keyword extraction, and topic clustering pipeline for automated content understanding.
- Evaluated the system on 31 diverse documents (PDF, image, text), achieving a 100% processing success rate with an average end-to-end latency of 55.6s while supporting documents up to 2,800+ words.
- Integrated OCR, HuggingFace embeddings, Chroma vector storage, and local LLM inference to generate simplified educational explanations.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, FastAPI, pandas, NumPy, PyTorch, TensorFlow

AI/ML & LLM Stack: Transformers, NLP, Prompt Engineering, RAG, Vector Databases, OpenAI API, GROQ API, LangChain, LangGraph, Vapi, Deepgram

Developer Tools: Git, Docker, GitHub, VS Code, Postman, MongoDB, Firebase, Upstash

Coding Profiles: 350+ [LeetCode](#) problems solved

Concepts: REST APIs, Authentication, System Design Basics, Multi-Agent Systems, Data Structures & Algorithms

RELEVANT COURSEWORK

Data Structures and Algorithms | Object-Oriented Programming | Operating Systems | Database Management Systems | Computer Networks | Theory of Computation | Compiler Design | Artificial Intelligence and Machine Learning | Software Engineering | Discrete Mathematics